Exploring the Impact of Data Decay on the Efficacy of Machine Learning Models for Poverty Prediction

Abstract

Effectively targeting social welfare poses a major challenge for most low- and middle-income countries. In response, they commonly deploy proxy means tests, a rudimentary form of machine learning, used to predict socio-economic status based on household characteristics. In this context, little research has been dedicated to how data decay (i.e., outdated training data) inhibits the efficacy of a given model. The following work leverages three waves of survey data used by Rwanda for proxy means testing to evaluate how the use of outdated training data impacts a model's ability to identify low-income households. Our simulations – conducted for five different machine learning models – reveal considerable data decay, with the annual decrease in targeting performance ranging from 4 to 6 percentage points. After conducting this evaluation, we implement an unsupervised domain adaptation strategy to mitigate the impacts of data decay. Although this extension to the traditional approach yields marginal improvements in accuracy, we conclude that more consistent surveying is required to sustain the efficacy and equity of proxy means testing as an approach to poverty alleviation.

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1 Introduction

Social security programs are designed to protect low-income households from economic shocks, encourage investment in human capital, and ensure a decent standard of living across the lifecycle. In countries where the majority of households report their income through tax filings, governments are well-equipped to identify individuals who are socio-economically disadvantaged. But most low- and middle-income countries face a dual challenge: limited or incomplete social registries and a large share of the workforce employed in the informal sector (Ohlenburg et al., 2022). These constraints make it significantly more difficult to target social welfare.

In response to this challenge, Grosh and Baker (1995) proposed using proxy means tests (PMTs). The method that has since been adopted by several countries in South Asia and Sub-Saharan Africa (Aiken et al., 2023). Proxy means tests rely on two types of survey data. The first is a nationally representative household survey with detailed information on demographic characteristics, asset ownership, and consumption expenditure. Since the survey contains data on attributes (i.e., covariates) that could be indicative of 'poverty' as well as actual 'labels' on consumption expenditure (a proxy for socio-economic status), it can be used to train machine learning models that can predict poverty status from observable household characteristics. But given the high cost of conducting these comprehensive surveys, they are typically carried out infrequently for relatively small samples (Brown et al., 2018). The second survey consists of more current, large-scale data on observable features, typically conducted via a shorter, cost-effective questionnaire that excludes information on consumption or income. Features in this survey are then used to determine eligibility for social security programs based on predictions from the model trained on the baseline survey.

Proxy means tests are used in more than fifty low- and middle-income countries, collectively representing more than a billion people, making them one of the most consequential applications of machine learning in public policy (Aiken et al., 2023). The literature exploring the efficacy of these approaches is extensive, but it generally assumes that the model used for the proxy means test is trained and applied at the same point in time. In practice, this premise rarely holds. As noted by Barca and Hebbar (2021) low- and middle-income countries tend to conduct survey type 1 every five to eight years. This could lead to significant discrepancies between the training data and the target data.

We leverage three waves of survey data collected by Rwanda's National Institute of Statistics in 2010, 2013 and 2016 to train five different models and evaluate their performance on subsequent waves of the survey, thereby simulating how data decay impacts the efficacy of a proxy means testing. The findings reveal that all models – including Ordinary Least Squares regression, Stepwise regression, LASSO regression, random forest, and neural network – suffer considerably from data decay. We then propose an *unsupervised domain adaptation strategy* that attributes weights to samples that more closely resemble the distribution of the target population during the training phase. Although this method leads to minor improvements, the study makes it clear that more consistent surveying or alternative methods for targeting are required to uphold the effectiveness of social protection programs.

2 Related work

The performance of proxy means testing has been studied extensively in the econometrics literature. Within this discourse there is an emerging consensus that although proxy means tests are imperfect, they generally perform better than alternative approaches (Alatas et al., 2012; Premand and Schnitzer, 2021; Hossain et al., 2025). Most of these works evaluate proxy means tests under the premise that training data and real-world data will have similar distributions, but in dynamic socio-economic environments new distributions invariably emerge and potentially inhibit the reliability of predictions. This issue is formally known as '*dataset shift*' in the literature on machine learning. Broadly speaking, there are three types of dataset shift:

- **Covariate Shift:** Occurs when the distribution of input features differs between the training and test sets, while the conditional distribution of the target given the inputs remains the same. This can arise due to sampling bias, non-random selection, or a changing environment.
- **Label Shift:** Refers to a change in the distribution of the output variable (i.e., labels or target classes) across training and testing data. This often occurs when certain outcomes are more or less prevalent in the target than in the source domain.
- **Concept Shift:** Describes a change in the underlying relationship between inputs and outputs. That is, the functional mapping from features to labels differs across domains, making it more difficult for models trained on past data to generalize accurately (Rabanser et al., 2019).

Intuitively, we expect that covariate shifts and concept shifts will be prevalent within survey data. To this end, we extend frameworks typically used to quantify dataset shift to demonstrate that applications of proxy means testing that do not take into account data decay are insufficient and misleading. Considerable literature has also been dedicated to mitigating dataset shift. Popular methods include domain adaptation (Zhang et al., 2013), adaptive learning (Bifet and Gavalda, 2009) and transfer learning (Zhang et al., 2019). To our knowledge, there are no works that have leveraged these tools for proxy means tests. For the purposes of this work, we merely focus on one of these techniques. Specifically, we follow Sugiyama et al. (2008) by deploying a logistic regression classifier to estimate the likelihood that a given observation originates from the test domain as opposed to the training domain. These probabilities are then used to construct weights that amplify the influence of training samples that more closely resemble the target population.

3 Methodology

3.1 Data

Three waves of publicly available survey data from 2010, 2013, and 2016 were obtained from Rwanda's National Institute of Statistics. The surveys, titled "Integrated Household Living Conditions Survey," contain data on over 400 attributes for roughly 14,000 households in each wave; some of these are provided at the household level, whilst others are available for all members of a household (National Institute of Statistics of Rwanda, 2025). It should be noted that the 'outdated' nature of the data does not take away from the study's validity

since we merely focus on quantifying expected data decay by simulating proxy means tests. Our results remain informative given that (a) Rwanda still uses the same survey to conduct proxy means testing and (b) the features included in these surveys are relatively similar in all low- and middle-income countries (Aiken et al., 2023). More recent survey data is not available for Rwanda since no survey was conducted during the COVID-19 pandemic, whilst other countries that leverage the tool rarely disclose the full list of survey covariates used to calibrate their models.

Three types of attributes were selected for the evaluation. Namely, demographic information on the head of household (e.g., sex, marital status, literacy, health insurance and level of education), data on asset-ownership (e.g., internet connection, mobile telephone, value of last harvest) and features related to housing (e.g., occupancy status, roofing material and mode of rubbish disposal). A list of all 40 covariates used can be found in Appendix A. All variables were first harmonized since spelling and phrasing varied across the three survey waves. Rule-based imputation was then used to address missing values where possible. Finally, continuous variables were winsorized at a 99% limit and subsequently standardized to a $[0, 1]$ range. The remaining categorical features were either one-hot encoded or ordinally encoded – more detailed information on this can be found in Appendix A. The target variable (i.e., label) was defined as *log transformed equivalenced consumption* to account for household size and composition, as is traditionally the case for proxy means testing.

3.2 Experimental setup

Since Areias and Wai-Poi (2022) have found that no machine learning model works best across the board for proxy means testing, we focus on evaluating data decay for five of the most popular ones – each currently deployed in at least one country to target social welfare (Aiken et al., 2023). We also evaluate a neural network, given increasing applications of deep learning to tabular data (Arik and Pfister, 2021; Katzir et al., 2021). To train these models, each wave is randomly split into a training set containing 75% of households and a test set containing 25% of households. The five models are then trained to predict log-transformed equivalenced consumption for each wave. To mitigate for idiosyncrasies in random data splits, the simulation is repeated 100 times; we then derive the average for each evaluative metric. This results in the following train-test configurations:

- Benchmark (within-wave performance):
  - Train on 2010 $\rightarrow$ Test on 2010
  - Train on 2013 $\rightarrow$ Test on 2013
- Exploring data decay:
  - Train on 2010 $\rightarrow$ Test on 2013
  - Train on 2010 $\rightarrow$ Test on 2016
  - Train on 2013 $\rightarrow$ Test on 2016

3.3 Evaluation metrics

To assess performance we evaluate the *Spearman's rank* correlation between the predicted ranking and the ground truth ranking. Additionally, we measure the *Targeting Error Rate* (TER) for a hypothetical social security program designed to reach the poorest 20% of households. Following Brown et al. (2018) we take a quota approach to calculating the TER. Specifically, an *exclusion error* occurs when a household is among the poorest in the ground truth but not in the predicted ranking, and an *inclusion error* occurs when a household is falsely predicted to be in the poorest 20%. Since the number of households selected is fixed by design, the inclusion and exclusion error rates are necessarily equal here.

3.4 Machine learning models

3.4.1 Parametric models

Traditional proxy means tests estimates household consumption as a weighted function of observable features x_{it}. The most common approach uses a linear regression, where *log consumption* is regressed on a set of household-level covariates. The estimated coefficients then serve as weights in the scoring function. Formally, for a survey sample of size N, we estimate the following empirical model:

$$\log(y_{it}) = \alpha + x'_{it}\beta + \epsilon_{it}. \tag{1}$$

Here y_{it} denotes consumption for household i at time t, x'_{it} is a vector of observed covariates, and β is a vector of corresponding coefficients used for prediction. This constitutes a simple **Ordinary Least Squares (OLS) regression**. Next, we implement a **Stepwise Regression**[1] to select a sparse subset of covariates – one of the most commonly deployed approaches to proxy means testing (Aiken et al., 2023). The procedure is applied separately for each random train-evaluation split as follows: First, the training set is further divided into a selection set and a calibration set. This stepwise selection method maintains the same linear form as Equation (1), but restricts the vector x_{it} exclusively to covariates chosen by a greedy linear model fit to the selection set, where covariates are iteratively added based on how they contribute to the calibration set. This approach takes inspiration from Haushofer and Shapiro (2018); variables continue to be added until the mean squared error on the calibration set no longer improves. The final selected set of covariates is then used to retrain the linear model on the full training data before performance is evaluated on the test set.

Additionally, we evaluate the performance of a **LASSO regression**, which also assumes a linear functional form as specified in Equation (1), but introduces an ℓ_1 penalty term that shrinks some coefficients to zero. This enables simultaneous regularization and variable selection, potentially allowing for a more stable alternative to stepwise methods. LASSO has gained traction in recent applications of proxy means testing since it is particularly useful in high-dimensional settings, or when many covariates are only weakly predictive of the outcome(Ligon and Trachtman, 2024). Formally, LASSO estimates the vector of coefficients β by solving the following optimization problem:

[1]For computational efficiency, three adjustments were made here: (i) the maximum number of covariates included in the model was capped at 15; (ii) features with zero variance or low absolute correlation (<0.01) were excluded during pre-screening; and (iii) the number of simulations was reduced to 10.

$$\min_{\beta} \left\{ \sum_{i=1}^{N} \left(y_{it} - x_{it}'\beta\right)^2 + \lambda \sum_{j=1}^{K} |\beta_j| \right\}, \tag{2}$$

where $\lambda \geq 0$ is the regularization parameter that controls the strength of penalization. Larger values of λ lead to sparser models by shrinking more coefficients to exactly zero. Following Hastie (2009) the regularization strength λ is selected via three-fold cross-validation, using only the training data to ensure that model selection remains out-of-sample with respect to the evaluation set.

3.4.2 Non-parametric models

Random forests are non-parametric ensemble learning methods that combine predictions made by multiple decision trees to produce a more robust and accurate model. In the context of proxy means testing, Random Forests can capture complex, non-linear relationships between household characteristics and consumption, which may be missed by traditional linear models. Formally, the predicted log-consumption for household i at time t is given by:

$$\hat{y}_{it}^{\text{RF}} = \frac{1}{B} \sum_{b=1}^{B} T_b(x_{it}), \tag{3}$$

where $T_b(\cdot)$ denotes the prediction of the b-th decision tree, and B is the total number of trees in the ensemble. Each tree is trained on a bootstrap sample of the data and a random subset of covariates, which helps reduce variance and avoid overfitting. Random forests are trained on the training set and evaluated on a held-out test set. The ensemble size (B) and maximum tree depth are chosen via three-fold cross-validation on the training set to optimize predictive performance. This process ensures a balance between predictive accuracy and model generalizability.

Recent studies have found that deep **neural networks** can outperform traditional classifiers on tabular data (Arik and Pfister, 2021; Katzir et al., 2021; Popov et al., 2020). Although neural networks are not currently deployed in any proxy means testing efforts (to our knowledge), they are worth evaluating in the context of these findings. Following Shwartz-Ziv and Armon's (2022) application of deep learning in survey data, we use random search to select hyperparameters for the model's parameters. Specifically, the following architecture was selected: a feedforward neural network with two hidden layers, the first consisting of 64 units and the second of 32 units, both using ReLU activation functions. The model was trained using the Adam optimizer to minimize mean squared error, with a batch size of 32, early stopping after 5 epochs of no improvement, and a maximum of 100 training epochs (the full search space explored during the random search is detailed in Appendix B).

Mathematically, we let $\mathbf{x} \in \mathbb{R}^d$ denote the input feature vector for a given household, and let $\hat{y} \in \mathbb{R}$ denote the predicted log-consumption. The neural network maps input features to output prediction through a sequence of affine transformations and non-linearities:

$$h_1 = \text{ReLU}(W_1 x + b_1) \tag{4}$$

$$h_2 = \text{ReLU}(W_2 h_1 + b_2) \tag{5}$$

$$\hat{y} = W_3 h_2 + b_3, \tag{6}$$

where $W_1 \in \mathbb{R}^{64\times d}$ and $b_1 \in \mathbb{R}^{64}$ define the weights and biases of the first hidden layer, which outputs a 64-dimensional representation. $W_2 \in \mathbb{R}^{32\times 64}$ and $b_2 \in \mathbb{R}^{32}$ define the second hidden layer, and $W_3 \in \mathbb{R}^{1\times 32}$ and $b_3 \in \mathbb{R}$ produce the scalar output prediction. The model is trained to minimize mean squared error loss:

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2. \tag{7}$$

3.5 Mitigating dataset shift

Following prior work on covariate shift correction by Sugiyama et al. (2008) and Ataei et al. (2021) we implement an unsupervised domain adaptation approach to improve a given model's robustness to temporally shifted surveys. The idea here is that there are four pieces to the targeting puzzle: (1) training data from previous years, (2) training labels from previous years, (3) training data from the current year, and (4) labels for the current year. In our context, labels from the current year are not available, but the third piece of the puzzle (i.e., the distribution of current covariates) can be used to improve how the model is trained, despite the absence of current labels.

Mathematically, the problem can be set up as follows. Let $\mathcal{D}_S = \{(x_i, y_i)\}_{i=1}^{n}$ denote the source domain consisting of labelled features from an older survey wave (e.g., 2010), where $x_i \in \mathbb{R}^m$ are feature vectors and y_i is log-consumption. Let $\mathcal{D}_T = \{x_j\}_{j=1}^{n'}$ denote the target domain from the current survey wave, which is assumed to be unlabelled. In this context we then adopt the standard covariate shift assumption:

$$P_S(y \mid x) = P_T(y \mid x) \quad \text{but} \quad P_S(x) \neq P_T(x). \tag{8}$$

Our goal here is to adjust the training procedure so that the model trained on $\mathcal{D}_S$ generalizes better to $\mathcal{D}_T$ by estimating the distributional shift between source and target covariates. To this end, we train a *domain classifier* to determine whether a feature vector x originates from the source or target domain. Specifically, we define a binary classification dataset and its corresponding domain labels d_i:

$$\mathcal{D}_{\text{dom}} = \{(x_i, d_i)\}_{i=1}^{n+n'}, \quad d_i = \begin{cases} 0 & \text{if } x_i \in \mathcal{D}_S \\ 1 & \text{if } x_i \in \mathcal{D}_T. \end{cases} \tag{9}$$

The following logistic regression is then fit to this dataset:

$$D(x) = P(d = 1 \mid x) = \sigma(\mathbf{w}^\top x + b), \tag{10}$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function applied to a linear transformation of the input features and $(\mathbf{w}, b)$ are parameters learned via the maximum likelihood estimation. The logistic loss for a sample (x_i, d_i) is:

$$L_{\text{logit}}(x_i, d_i) = d_i \log D(x_i) + (1 - d_i)\log(1 - D(x_i)). \tag{11}$$

The model is trained by minimizing average loss over the full domain-labelled dataset $\mathcal{D}_{\text{dom}}$. After training the logistic regression model, we compute *importance weights* for each source sample using the density ratio:

$$\beta(x) = \frac{P_T(x)}{P_S(x)} = \frac{D(x)}{1 - D(x)}. \tag{12}$$

This expression arises from Bayes' rule under equal prior class probabilities (i.e., $P(d = 0) = P(d = 1)$). The weight $\beta(x)$ increases for source examples that resemble the target data, and down-weights those that are unlikely under the target distribution, essentially revealing how 'target-like' each household is. Additionally, to ensure numerical stability, we apply clipping and normalization:

$$D_{\text{clipped}}(x) = \min(1 - \epsilon, \max(\epsilon, D(x))) \quad \text{with } \epsilon = 0.01 \tag{13}$$

$$\beta(x) = \frac{D_{\text{clipped}}(x)}{1 - D_{\text{clipped}}(x)} \tag{14}$$

$$\tilde{\beta}(x) = \frac{\beta(x)}{\frac{1}{n}\sum_{i=1}^{n} \beta(x_i)}. \tag{15}$$

By then training our downstream models using these derived weights we have adapted it to dataset shift in the target domain.

$$\hat{R}_{\text{weighted}}(f) = \frac{1}{n} \sum_{i=1}^{n} \tilde{\beta}(x_i) \cdot \ell(f(x_i), y_i), \tag{16}$$

where $\ell(f(x_i), y_i)$ is the standard loss used to train a given model. All weights $\tilde{\beta}(x_i)$ are passed via the `sample_weight` parameter in `scikit-learn` models.

4 Results

4.1 Dataset shift

Table 1: Spearman correlation across configurations by model

Configuration	OLS	Step-wise	LASSO	Random Forest	Neural Network
2010 → 2010	0.59	0.56	0.59	0.66	0.43
2010 → 2013	0.53	0.51	0.51	0.52	0.10
2010 → 2016	0.43	0.43	0.47	0.41	0.03
2013 → 2013	0.46	0.45	0.49	0.54	0.36
2013 → 2016	0.11	0.11	0.17	0.19	-0.14

Table 2: TER (20%) results across configurations by model

Configuration	**OLS**	**Stepwise**	**LASSO**	**Random Forest**	**Neural Network**
2010 → 2010	0.37	0.30	0.28	0.24	0.38
2010 → 2013	0.46	0.44	0.58	0.50	0.52
2010 → 2016	0.64	0.62	0.63	0.61	0.83
2013 → 2013	0.43	0.35	0.35	0.36	0.41
2013 → 2016	0.62	0.53	0.51	0.55	0.83

Table 1 and Table 2 offer insights into the observed level of dataset shift. Notably, there is clear data decay across all models. For the benchmark all five models perform better on the 2010 data than on the 2013 data, perhaps because the 2013 survey includes a smaller number of households. When looking at individual models, Random Forest and LASSO perform best with TERs of 0.24 and 0.28 on the 2010 benchmark; however, neither is robust to temporal shifts with the TER nearly doubling over the 6-year period for both models. The decay is also linear in most cases, indicating that there are gradual shifts in the covariate distribution, as opposed to sudden stepwise fluctuations (e.g., due to a natural disaster or a sudden influx of wealth). The estimated yearly decay based on these results for the TER would be between 4 and 6 percentage points for all models, except the neural network, which saw a sudden collapse in performance when deployed on temporally shifted data, with its Spearman correlation dropping to -0.14 for the 2013→2016 configuration. The collapse can most likely be attributed to the model overfitting due to insufficient fine-tuning.

4.2 Mitigating dataset shift

Table 3: Spearman correlation across configurations by model (with domain classifier)

Model	2010 → 2013		2010 → 2016		2013 → 2016	
	Normal	Adj.	Normal	Adj	Normal	Adj.
OLS	0.53	0.54	0.43	0.45	0.11	0.10
Stepwise	0.51	0.51	0.43	0.43	0.11	0.12
LASSO	0.51	0.54	0.47	0.49	0.17	0.17
Random Forest	0.52	0.54	0.41	0.44	0.19	0.20
Neural Network	0.10	0.07	0.03	0.01	-0.14	-0.14

Table 4: TER results across configurations by model (with domain classifier)

Model	2010 → 2013		2010 → 2016		2013 → 2016	
	Normal	Adj.	Normal	Adj.	Normal	Adj.
OLS	0.46	0.44	0.64	0.60	0.62	0.62
Stepwise	0.44	0.43	0.62	0.61	0.53	0.50
LASSO	0.58	0.55	0.63	0.60	0.51	0.52
Random Forest	0.50	0.48	0.61	0.58	0.55	0.53
Neural Network	0.49	0.49	0.83	0.81	0.83	0.83

The results presented in Tables 3 and 4 capture the extent to which the domain classifier mitigated for data decay. Table 3 shows that, for most models, applying the adjustment either improved or preserved the Spearman correlation between predicted and true household consumption ranks. The most consistent improvements are observed in the Random Forest and LASSO models, with correlations rising modestly from 0.52 to 0.54 and 0.47 to 0.49 respectively over the 2010→2016 configuration. These gains suggest that domain weighting helped the models better generalize to temporally shifted populations. For the OLS and stepwise regression, changes were marginal and inconsistent, whilst the neural network's performance showed mixed results, further highlighting its instability in this context.

5 Discussion

First, it should our models' performance on the benchmark is similar to what has been reported in the literature, confirming the validity of our hyper-parameters, whilst underscoring the generalizability of our findings. Our subsequent simulation makes it clear that although proxy means tests perform well in static environments, they fail under dataset shift. These results have considerable implications for social and economic policy. They suggest that when a the model underlying a proxy means test is three or more years out of date and retraining is not feasible, alternative targeting strategies most likely offer greater accuracy in the interim. Community-based targeting, for instance, is typically only marginally less accurate than proxy means testing (Karlan and Thuysbaert, 2019) and may serve as a practical substitute. Emerging digital data sources also offer promising alternatives. For example, Aiken et al. (2022) report a ten percentage point difference in targeting accuracy between a freshly calibrated proxy means test and a poverty index derived from mobile phone metadata. Simpler approaches such as geographic, categorical, or self-targeting may also be appropriate policy options when the ground truth required for a proxy means tests has not been recently updated.

This dataset shift can be better understood by taking a deep-dive into more granular feature distributions. For example, the number of people who owned a mobile phone doubled between 2010 and 2016. Similarly, there was a 30% increase in the number of people who reported knowing how to read and confidently use a computer. Finally, there was a considerable shift in the distribution of values for the "Distance to the nearest drinking water source (meters)" feature. These constitutes clear examples of *covariate shift*, where the distribution of input features differs across the training and test sets, whilst the condi-

tional distribution of the target remains relatively static. Notably, there were also *concept shifts* in the data. For example, Nyamagabe as a place of birth was associated with much greater poverty levels in 2010 than in 2016, likely due to the construction of a hydroelectric power station built in this district between 2010 and 2014 (Sharangabo and Szczepaniak, 2019). Understanding the likely causes behind the dataset shift allows for more informed variable selection during training phase as well as hyperparameter tuning that anticipates these changes.

Finally, the domain classifier used to reweigh features during the training phase led to minor but meaningful improvements in performance. Indeed, although the observed reductions in targeting error rates were small—ranging from 0 to 0.4 percentage points—they are significant in this context since proxy means tests are generally used to allocate welfare to +10,000 people; meaning that marginal gains in targeting accuracy can yield large positive social externalities. Domain classifiers should therefore be explored further, especially since the simple logistic regression model used here is typically outperformed by more advanced models (Rabanser et al., 2019).

6 Limitations and areas for future research

The work's primary limitation is that it merely considers three survey waves in one country. Generalizing the above-presented findings would require evaluating data decay with reference to additional survey data – ideally conducted on a yearly basis – in other countries. The second limitation is that the analysis includes merely 40 covariates with a strong focus on housing-related variables. Perhaps a greater emphasis on asset ownership or demographics would yield different results. Exploring how data decay varies for different covariates could provide valuable insights into what attributes should be selected to strengthen a model's robustness to data decay. The third limitation is that this study – alongside most of the literature on proxy means testing – disregards fairness and equity considerations. Since the test directly influence who receives social benefits, fairness should be seen as a key factor in their evaluation. A growing body of research in this regard has evolved through the works of Noriega-Campero et al. (2020) who explore subgroup fairness in targeting by measuring accuracy across population segments to avoid systematic disparities in eligibility.

7 Conclusion

In summary, this work's contributions are twofold. First, it quantifies how dataset shift impacts the efficacy of various machine learning models commonly deployed for proxy means testing. Our findings make it clear that acknowledging and addressing dataset shift is a necessary condition for responsible and equitable program design in this context. Second, it demonstrates that minor accuracy gains can be achieved when taking the target distribution into consideration during the training phase. As such, this study provides both an empirical assessment of data decay in Rwanda and a framework that policymakers can adopt to quantify and mitigate dataset shift when deploying proxy means tests in their respective context.

References

Aiken, E., Bellue, S., Karlan, D., Udry, C., & Blumenstock, J. E. (2022). Machine learning and phone data can improve targeting of humanitarian aid. *Nature*, *603*(7903), 864–870.

Aiken, E., Ohlenburg, T., & Blumenstock, J. (2023). Moving targets: When does a poverty prediction model need to be updated? *Proceedings of the 6th ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies*, 117–117.

Alatas, V., Banerjee, A., Hanna, R., Olken, B. A., & Tobias, J. (2012). Targeting the poor: Evidence from a field experiment in indonesia. *American Economic Review*, *102*(4), 1206–1240.

Areias, A., & Wai-Poi, M. (2022). Machine learning and prediction of beneficiary eligibility for social protection programs. *Revisiting Targeting in Social Assistance*, *507*.

Arik, S. Ö., & Pfister, T. (2021). Tabnet: Attentive interpretable tabular learning. *Proceedings of the AAAI conference on artificial intelligence*, *35*(8), 6679–6687.

Ataei, M., Erdogdu, M., Kocak, S. A., Ben-David, S., & Saleh, S. (2021). *Understanding dataset shift and potential remedies: A vector institute industry collaborative project* (tech. rep.). Vector Institute. https://vectorinstitute.ai/wp-content/uploads/2021/08/Understanding-Dataset-Shift-and-Potential-Remedies_Vector-Institute_2021.pdf

Barca, V., & Hebbar, M. (2021). 24. delivering social transfers. *Handbook on social protection systems*, 426.

Bifet, A., & Gavalda, R. (2009). Adaptive learning from evolving data streams. *International symposium on intelligent data analysis*, 249–260.

Brown, C., Ravallion, M., & Van de Walle, D. (2018). A poor means test? econometric targeting in africa. *Journal of Development Economics*, *134*, 109–124.

Grosh, M., & Baker, J. L. (1995). Proxy means tests for targeting social programs. *Living standards measurement study working paper*, *118*, 1–49.

Hastie, T. (2009). The elements of statistical learning: Data mining, inference, and prediction.

Haushofer, J., & Shapiro, J. (2018). The long-term impact of unconditional cash transfers: Experimental evidence from kenya. *Busara Center for Behavioral Economics, Nairobi, Kenya*, *10*.

Hossain, M., Jäckering, L., Mullally, C., & Winters, P. (2025). Poverty prediction and targeting over time and space: Evidence from nigeria. *Applied Economic Perspectives and Policy*.

Karlan, D., & Thuysbaert, B. (2019). Targeting ultra-poor households in honduras and peru. *The World Bank Economic Review*, *33*(1), 63–94.

Katzir, L., Elidan, G., & El-Yaniv, R. (2021). Net-{dnf}: Effective deep modeling of tabular data. *International Conference on Learning Representations*. https://openreview.net/forum?id=73WTGs96kho

Ligon, E., & Trachtman, C. (2024). Assessing targeting peformance: The case of ghana's leap program.

National Institute of Statistics of Rwanda. (2025). Integrated Household Living Conditions Survey (EICV).

Noriega-Campero, A., Garcia-Bulle, B., Cantu, L. F., Bakker, M. A., Tejerina, L., & Pentland, A. (2020). Algorithmic targeting of social policies: Fairness, accuracy, and distributed governance. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 241–251.

Ohlenburg, T., Pinxten, J., Fricke, D., & Caccioli, F. (2022). Fewer questions, more answers: Truncated early stopping for proxy means testing. *More Answers: Truncated Early Stopping for Proxy Means Testing (December 7, 2022)*.

Popov, S., Morozov, S., & Babenko, A. (2020). Neural oblivious decision ensembles for deep learning on tabular data. *International Conference on Learning Representations*. https://openreview.net/forum?id=r1eiu2VtwH

Premand, P., & Schnitzer, P. (2021). Efficiency, legitimacy, and impacts of targeting methods: Evidence from an experiment in niger. *The World Bank Economic Review*, *35*(4), 892–920.

Rabanser, S., Günnemann, S., & Lipton, Z. (2019). Failing loudly: An empirical study of methods for detecting dataset shift. *Advances in Neural Information Processing Systems*, *32*.

Sharangabo, D., & Szczepaniak, M. (2019). Economic development in rwanda–institutional perspective. *Zeszyty Naukowe Wyższej Szkoły Gospodarki w Bydgoszczy. Seria Ekonomia*, *10*(34), 107–124.

Sugiyama, M., Suzuki, T., Nakajima, S., Kashima, H., Von Bünau, P., & Kawanabe, M. (2008). Direct importance estimation for covariate shift adaptation. *Annals of the Institute of Statistical Mathematics*, *60*, 699–746.

Zhang, J., Li, W., Ogunbona, P., & Xu, D. (2019). Recent advances in transfer learning for cross-dataset visual recognition: A problem-oriented perspective. *ACM Computing Surveys (CSUR)*, *52*(1), 1–38.

Zhang, K., Schölkopf, B., Muandet, K., & Wang, Z. (2013). Domain adaptation under target and conditional shift. *International conference on machine learning*, 819–827.

A Features

Table 5: Demographic variables (applies to the head of household)

Variable	**Type**	**Operation**
Sex	Binary	Binary encoding
Age (Years)	Discrete	None
Marital status	Categorical	One hot encoding
Place of birth	Categorical	One hot encoding
Ever been to school	Binary	Binary encoding
Highest class succesfully attained in school	Categorical	Ordinal encoding
Able to read a simple written note	Binary	Binary encoding
Able to write a simple letter	Binary	Binary encoding
Able to do a written calculation	Binary	Binary encoding
Confidence to use computer	Categorical	Ordinal encoding
Health insurance	Categorical	One hot encoding
Household size	Discrete	None
Number of kids	Discrete	None

Table 6: Housing related variables

Variable	**Type**	**Operation**
Urban/rural 2002	Binary	Binary encoding
Type of habitat	Categorical	One hot encoding
Type of dwelling	Categorical	One hot encoding
Number of rooms occupied by household	Discrete	None
Number of bed rooms for household	Discrete	None
Owner of the dwelling	Categorical	One hot encoding
Current occupancy status	Categorical	One hot encoding
Main source of drinking water	Categorical	One hot encoding
Distance to the nearest drinking water source	Continuous	Winsorized/standardized
Amount paid to private water vendor last 7 days	Continuous	Winsorized/standardized
Main source of home lighting	Categorical	One hot encoding
Amount paid on electricity in last 4 weeks	Continuous	Winsorized/standardized
Primary source of cooking fuel	Categorical	One hot encoding
Mode of rubbish disposal	Categorical	One hot encoding
Type of toilet household use	Categorical	One hot encoding
Main source of info on environmental issues	Categorical	One hot encoding
Main construction material of exterior wall	Categorical	One hot encoding
Main roofing material for dwelling	Categorical	One hot encoding
Main floor material for dwelling	Categorical	One hot encoding

Table 7: Asset Ownership variables

Variable	Type	Operation
Has a mobile telephone	Binary	Binary encoding
Internet connection at home	Binary	Binary encoding
Household reported owning any land	Binary	Binary encoding
Total surface area of land owned	Continuous	Winsorized/standardized
Household harvested large crops in last 12 months	Binary	Binary encoding
Total value of large crop harvest	Continuous	Winsorized/standardized
Household raised any livestock in last 12 months	Binary	Binary encoding
Total value of livestock	Continuous	Winsorized/standardized

B Hyperparameter Search Space

Random Forest

- Number of trees (n_estimators): Discrete choice {100, 200, 300}
- Maximum tree depth (max_depth): Discrete choice {5, 10, 15, None}
- Minimum sample split (min_samples_split): Discrete choice {2, 5}
- Minimum samples per leaf (min_samples_leaf): Discrete choice {1, 2}
- Scoring metric: Negative mean squared error (MSE)
- Search method: RandomizedSearchCV with 3-fold cross-validation

Neural Network

- Number of hidden layers: Discrete choice {1, 2, 3, 4}
- Hidden layer sizes: Discrete choice {16, 32, 64, 128}
- Activation function: Categorical choice {ReLU, Tanh, LeakyReLU}
- Optimizer: Categorical choice {Adam, RMSprop, SGD}
- Learning rate: Log-uniform distribution $\in [10^{-4}, 10^{-2}]$
- Batch size: Discrete uniform choice {16, 32, 64, 128}
- Early stopping patience: Fixed at 5 epochs
- Maximum number of epochs: Fixed at 100
- Loss function: Mean squared error (MSE)